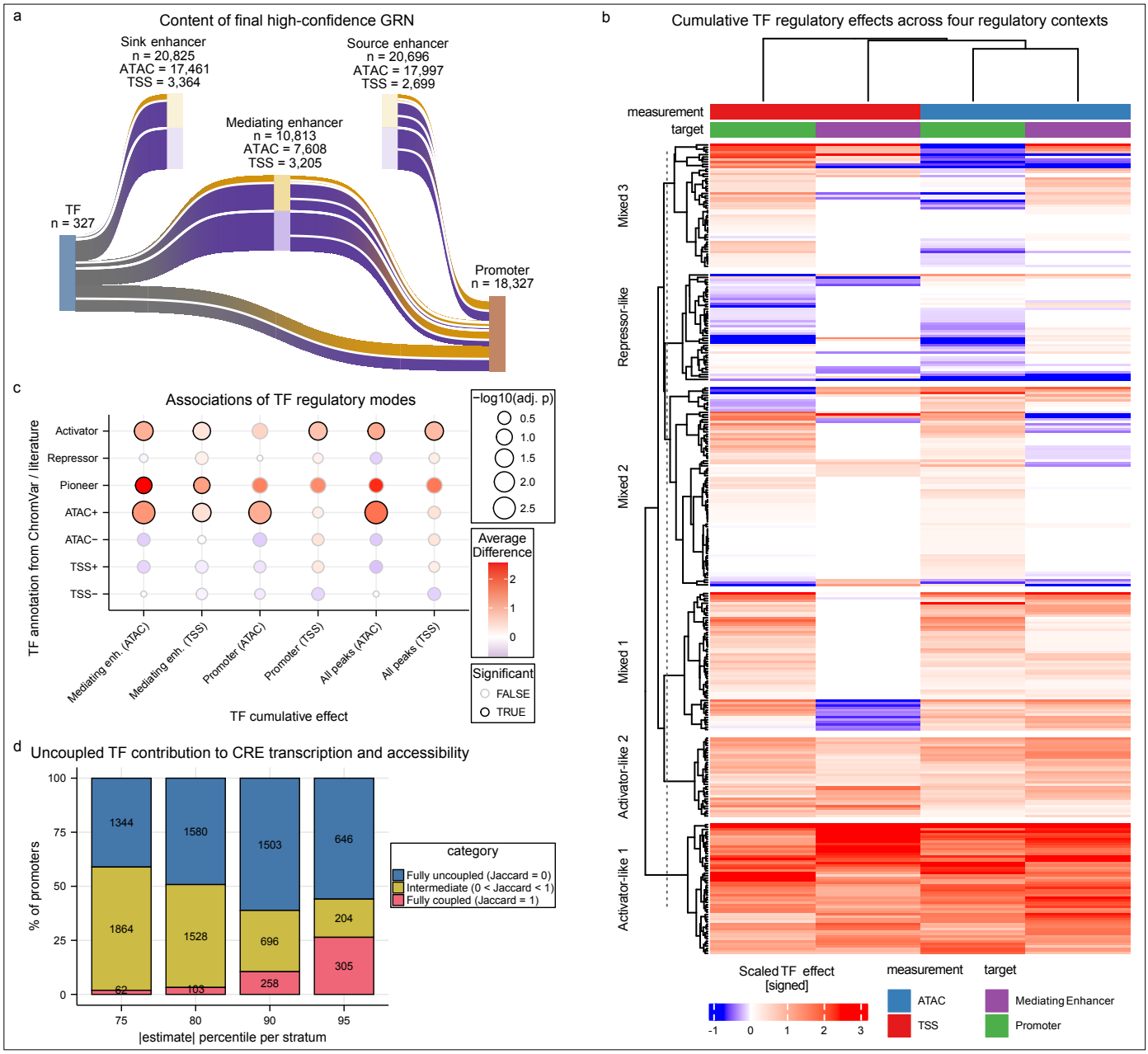


Figure 5 | Structure and transcription factor regulatory modes in the gene regulatory network.

a, Sankey diagram of the final high-confidence GRN, filtered by adjusted p-value ≤ 0.05, deviance ratio ≥ 0.3, per-stratum |coefficient| ≥ 90th percentile. Nodes represent TFs, sink enhancers (TF-regulated without promoter connection), mediating enhancers (TF-regulated and connected to promoters), source enhancers (connected to promoters without TF regulation) and promoters. Ribbon width represents log-transformed edge counts and color indicates the regulatory modality (violet, chromatin accessibility [ATAC]; gold, transcription [TSS]; grey, TF expression). Cross-modality edges transition between the corresponding colors. The network contains 31,130 direct TF–promoter edges and, through mediating enhancers, 14,221 TF–enhancer and 23,695 enhancer–promoter edges. **b**, Heatmap of scaled, signed cumulative TF regulatory effects across four regulatory contexts: mediating-enhancer accessibility (ATAC), mediating-enhancer transcription (TSS), promoter accessibility and promoter transcription. Rows represent TFs and were hierarchically clustered into six groups according to their regulatory profiles. Red and blue indicate positive and negative cumulative effects, respectively. **c**, Associations between GRN-derived cumulative TF effects (columns) and TF functional annotations derived from ChromVar analysis and literature curation (rows). Dot size represents $-\log_{10}$(BH-adjusted *p*-value) from linear-model tests, and color represents the difference in cumulative regulatory effect between TFs with and without each annotation (red, positive; blue, negative). Black-outlined dots indicate significant associations (BH-adjusted $p < 0.1$). **d**, Overlap between TF sets associated with promoter accessibility and promoter transcription across GRN edge-weight thresholds. Each bar represents one percentile threshold of absolute edge coefficients, where the 90th-percentile threshold corresponds to the final high-confidence GRN. Only promoters with at least one TF association in both modalities are included. Colors indicate whether accessibility- and transcription-associated TF sets are completely distinct, partially overlapping or completely overlapping.